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traveltime: an R package to calculate travel time across a landscape from user-specified locations

Gerard E. Ryan [1,2¶], **Nicholas Tierney** [1,3], **Nick Golding** [1,4], and **Daniel J. Weiss** [1,3]

1 The Kids Research Institute Australia, Nedlands 6009 WA, Australia **2** Melbourne School of Population and Global Health, University of Melbourne, 3010, VIC, Australia **3** Curtin University, Bentley, WA, Australia **4** University of Western Australia, WA, Australia ¶ Corresponding author

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Software

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Summary

Understanding and mapping the time to travel among locations is useful for many activities from urban planning (Zahavi, 1974) to public health (Hulland et al., 2019; Weiss et al., 2020) and myriad others (Nelson et al., 2019). Here we present a software package — `traveltime` — written in and for the language R (R Core Team, 2024). `traveltime` enables a user to create a raster map of the travel time over an area of interest from a user-specified set of geographic coordinates. Applications include: estimating sampling bias (Dennis & Thomas, 2000), allocating defibrillators (Tierney et al., 2018), setting health districts (Padgham et al., 2019), or mapping access to vehicle chargers (Falchetta & Noussan, 2021) and agricultural facilities (Zhao et al., 2023).

The work-flow requires two key steps:

- preparing a friction surface for the area of interest, and then
- calculating travel time over that surface for the points of interest.

`traveltime` provides a spatial interface using object classes from the `terra` package (Hijmans, 2024). Travel time is calculated as movement over a resistance 'friction surface' (van Etten, 2017). To provide easy access to the existing friction surfaces generated by Weiss et al. (2020), `traveltime` internally uses `malariaAtlas` (Pfeffer et al., 2018) to download surfaces for the area of interest; though users can also supply any other friction surface.

`traveltime` is available from R-Universe and GitHub, and documented at https://idem-lab.github.io/traveltime/. Although this article is intended to be the key reference for the `traveltime` package, we suggest citations of the package are accompanied by citing the underlying methodological work as well (Weiss et al., 2018, 2020).

Statement of need

Global maps of travel time to cities (Nelson et al., 2019; Weiss et al., 2018) and health care (Hulland et al., 2019; Weiss et al., 2020) have generated much interest and use[1], and the city data set of Nelson et al. (2019) is available to R users through the widely-used `geodata` package (Hijmans et al., 2024). Here we extend that work to enable travel time calculations from any arbitrary set of locations and friction surface.

A gaggle of R packages provide superficially similar though fundamentally different functionality via the TravelTime.com API (Lo Russo, 2024; TravelTime, 2024a, 2024b; von Bergmann, 2024). Their 'Isochron' polygons — areas reachable within a given time from a given location — are

[1] Collectively >1600 citations per Google Scholar at the 28th of January 2025.

most comparable to what `traveltime::calculate_travel_time()` calculates. However, each isochron is a single polygon calculated is for a single point location and specified maximum travel time, rather than a raster of temporal gradation across a landscape, jointly for an arbitrary number of points, as in `traveltime`.

With `traveltime`, we provide free and open source software to estimate travel time from any number of user-supplied locations, across a complete area of interest, and with convenient access to motorised transport or walking friction surfaces with global coverage.

Example: walking from public transport in Singapore

Here we provide a very brief example to calculate and map the walking travel time from rail transport stations across Singapore. Complete code for this example is available as a vignettte in package documentation.

Prepare data and friction surface

We need two items of data:

- our area of interest — Singapore, and
- our points to calculate travel time from — Singapore's Mass Rapid Transit (MRT) and Light Rail Transit (LRT) stations.

We download `singapore_shapefile`, a polygon boundary of Singapore from the GADM (GADM, 2022) database using the `geodata` package (Hijmans et al., 2024) as our area of interest. Our points are the `traveltime::stations` data, containing coordinates of all LRT and MRT station exits in Singapore (Land Transport Authority, 2019). We then download a walking travel friction surface for Singapore:

```
friction_singapore <- traveltime::get_friction_surface(
    surface = "walk2020",
    extent = singapore_shapefile
  )|>
  terra::mask(singapore_shapefile)
```

(Alternatively, we could use `surface = "motor2020"` for motorised travel. We're also only interested in walking *on land* so we `mask` out areas outside of the land boundary in `singapore_shapefile`)

We plot these data below. `traveltime` takes resistance values of friction (van Etten, 2017), so higher values of friction indicate more time travelling across a given cell.

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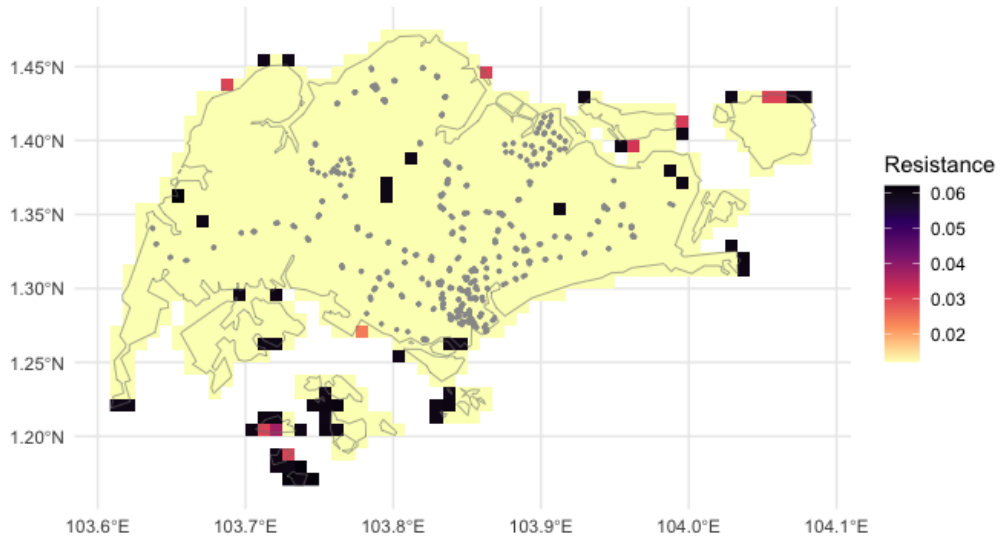

Figure 1: Friction surface raster of Singapore, showing Singapore boundary in grey, and station locations as grey points.

Calculate and plot the travel time

With all the data collected, the function `calculate_travel_time()` takes the friction surface `friction_singapore` and the points of interest in `stations`, and returns a `SpatRaster` of walking time in minutes to each cell from the nearest station:

```
trave_time_singapore <- calculate_travel_time(
  friction_surface = friction_singapore,
  points = stations
)

trave_time_singapore
```

```
class       : SpatRaster
dimensions  : 37, 57, 1  (nrow, ncol, nlyr)
resolution  : 0.008333333, 0.008333333  (x, y)
extent      : 103.6083, 104.0833, 1.166667, 1.475  (xmin, xmax, ymin, ymax)
coord. ref. :
source(s)   : memory
name        : travel_time
min value   :           0
max value   :         Inf
```

Resulting travel times are Figure 2[2].

[2]In Figure 1, the islands to the south and north-east are filled, but unconnected with the mainland. These cells appear absent in Figure 2; because they are not connected to any cells with a station, the calculated travel time is infinite, and so are not plotted in Figure 2.

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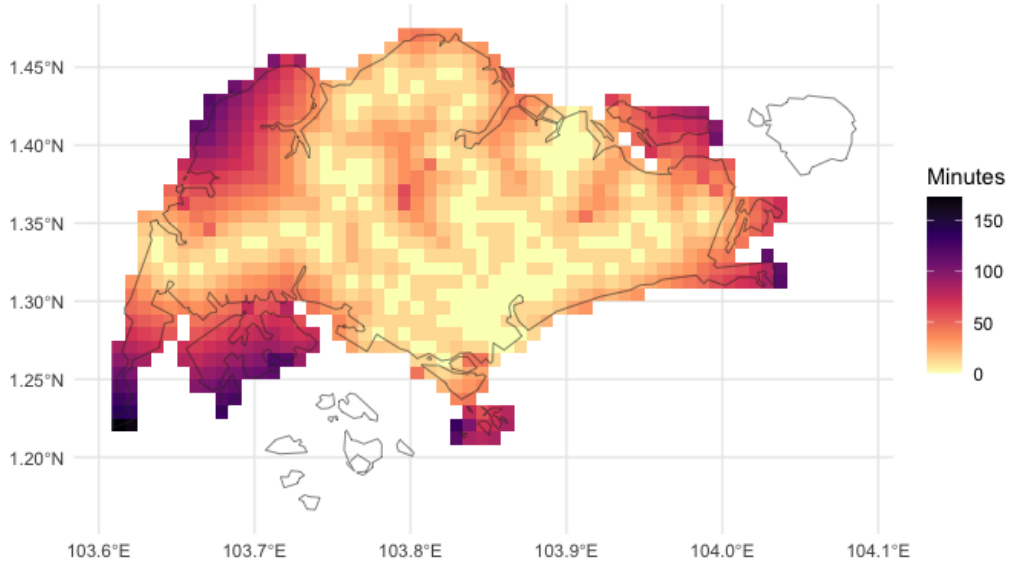

Figure 2: Map of walking travel time in Singapore, in minutes from nearest MRT or LRT station.

Opportunities for future development

The `traveltime` package is immediately suitable to a range of application. Nonetheless, we see opportunities to build the package utility through:

- capability to better distribute a wider range friction surfaces, and
- additional methods to efficiently compute results over large spatial extents.

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